Antranig Baghdassarian

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Education

Cornell University, Ithaca, NY – Bachelor of Science in Biomedical Engineering — Expected May 2027

- Emphasis in Electrical Engineering | Certifications in LEAN Yellow Belt & Engineering Leadership

Work Experience

Mechatronics Engineering Intern, LEADOPTIK - San Jose, CA — May 2026 – Aug 2026

- Develop functional test rig for main printed circuit board (PCB) to streamline board-level testing during manufacturing
- Design and 3D-print a catheter evaluation test fixture, streamlining test setup; improving throughput and repeatability

Quality & Compliance Intern, Johnson & Johnson Vision – Jacksonville, FL — May 2025 – Aug 2025

- Streamlined manual process flow via Tableau thus decreasing backorder occurrence and increasing visibility of stock
- Prototyped an automated product crate validation process with Power Automate via GxP tools, decreasing occurrence of human error & projected savings of 9+ hours/month

Undergraduate Researcher (REU), Tamerler Lab, University of Kansas – Lawrence, KS — May 2024 – July 2024

- Led 8 independent bacterial culture experiments via assays to evaluate effectiveness of novel bi-functional peptide
- Analyzed cell viability and collagen fibril formation via Cytation software and atomic force microscopy respectively
- Presented results of bi-functional peptide to maintain antimicrobial and collagen forming properties at symposium

Technical Experience

Student Researcher/Research Assistant, Dr. Edwin Kan's Lab - Ithaca, NY — Jan 2026 – Present

- Assist and carry out experimental design for non-invasive blood flow monitoring through brain via 900MHz
- Define experimental setups and protocols, evaluating antenna placement, sensor fixation, and physiological conditions including breath holding and Valsalva maneuvers via calibration with others.
- Develop a Python analysis workflow with mentor guidance to process synchronized RF and ECG recordings, segment cardiac cycles, and compare averaged pulse-waveform morphology across conditions.

Technical Project Manager, Empathera – Ithaca, NY — Mar 2024 – Present

- Prototype integrated muscle stimulation device to help prevent pressure ulcers and increase hospital's quality rating
- Ran finite elemental analysis simulations of stimulator placement, finding optimal location and power
- Won 5 awards in various business competitions, leading to +$10k in non-dilutive funding as featured in: Weill Medicine

Electrical Engineering Team Co-Lead, Nexus Project Team – Ithaca, NY — Feb 2024 – Jan 2026

- Oversee weekly team meetings, deliverables, cross-team talk, budget, and $100k+ Altium (PCB design) sponsorship
- Lead electrical integration of 7 devices onto rover while training 10 new members with Arduino kits and tailored lessons
- Analyze motor encoder signal then improve Python script to accurately read encoder data and rotate motors
- Co-design base station charging system to decrease rover downtime via relays and validated current charge controller

Leadership Experience

Events & Planning Chair, Cornell Maker Club – Ithaca, NY — Feb 2024 – Present

- Organize 20+ workshops (30+ people), annual makeathon, while utilizing event feedback and brainstorming new events

President, Biomedical Engineering Society – Ithaca, NY — Oct 2023 – Present

- Establish club's website, official university email, society membership & monitor emails, increasing visibility of club
- Lead meetings and take minutes, brainstorm and organize events while establishing interclub connections for longevity

Skills & Interests

Software: Fusion 360, SOLIDWORKS, KiCad, Ansys Fluent, Power Automate, Tableau, Bambu Studio, LabVIEW
Certifications: LEAN Yellow Belt, Leadership in Engineering Fields
Languages: Python, Java, MATLAB, C, R, Spanish | **Development:** Soldering, Programming with Microcontrollers
Interests: Electronics**,** Engineering How-To Videos, Running, Volleyball, My Hero Academia, and Kaizen Mentality